

Quantifiers

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Quantifier operations return a Boolean value that indicates whether some or all of the elements in a sequence satisfy a condition.

Method Name	Description
Any	Determines whether any elements in a sequence satisfy a condition.
All	Determines whether all the elements in a sequence satisfy a condition.
Contains	Determines whether a sequence contains a specified element.

Any:

بترجع True في حالة ان عنصر واحد بس حقق ال condition.

بتلف على كل العناصر لغاية ما تلاقي عنصر يحقق ال condition عشان كد ال time بتاعها $O(n)$.

All:

بترجع True فقط في حالة ان كل العناصر حققت ال condition.

بتلف على كل العناصر لغاية ما تلاقي عنصر مش بيققق ال condition عشان كد ال time بتاعها $O(n)$ برضو.

Contains:

`Contains` is a built-in method used to check if a specific value exists within a string or a collection, returning a Boolean (`true` or `false`).

عندي Instance Method و Extension Method.

1. Instance Method String.Contains:

بتشتغل على ال string

Used to check if a substring or character exists inside a larger string. By default, it is **case-sensitive**.

2. Extension Method `Collection.Contains` (Lists, Arrays, and LINQ)

هنا بتشتغل على `Collection`

Used to check if an exact item exists within collections like arrays, `List<T>`, or hash sets.

ال time بتاعها $O(n)$ لو بنشتغل على مجموعات زي `Arrays` او `Lists`

وبتكون $O(1)$ في حالة `Hash sets`

Feature	<code>String.Contains</code>	<code>Collection.Contains</code>
Purpose	Searches for partial matches (substrings).	Searches for exact matching elements.
Empty Value Behavior	<code>.Contains("")</code> always returns <code>true</code> .	Throws an exception or returns <code>false</code> depending on data types.